Digital Communications

OFDM

Orthogonal Frequency Division Multiplexing
2nd part

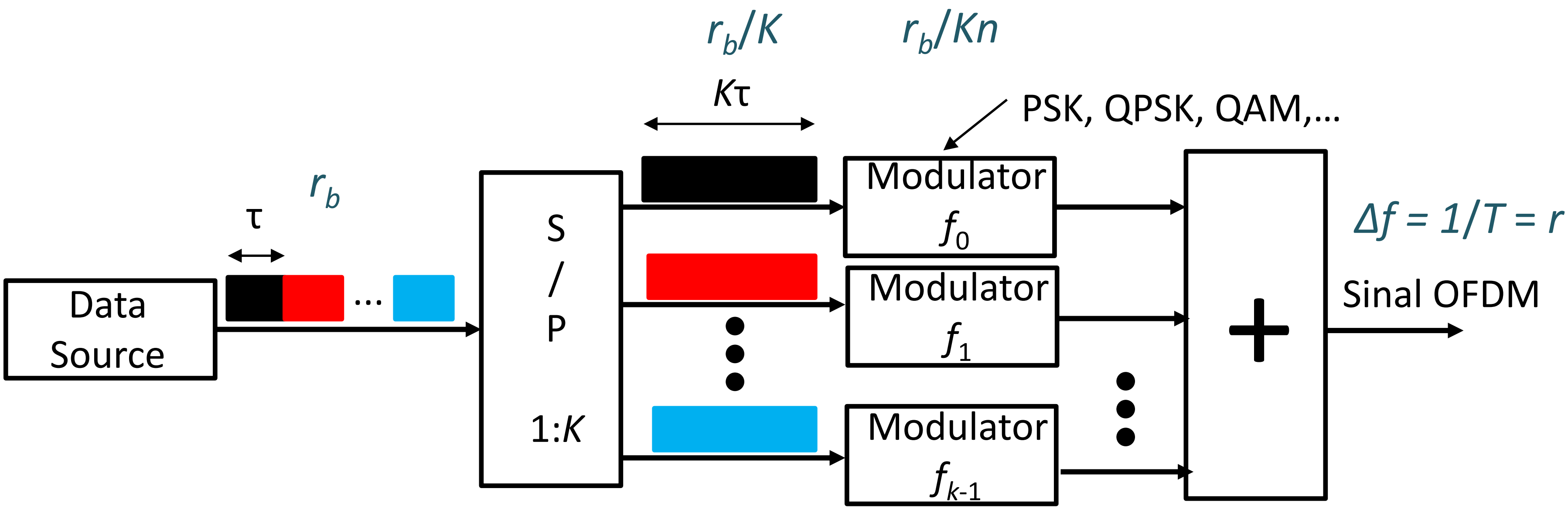
Universidade do Minho
Escola de Engenharia

Outline

- **OFDM – Orthogonal Frequency Division Multiplexing**
- OFDM Implementation
  - OFDM with the FFT
- Advantages of OFDM
- Disadvantages of OFDM
- Advanced Topics
- Applications

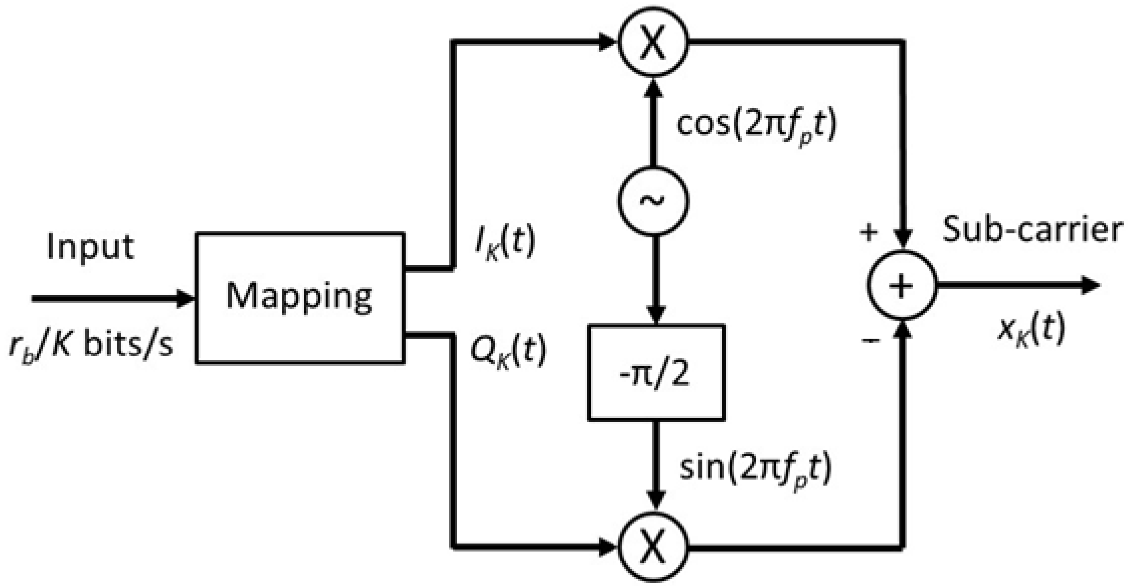
OFDM Implementation

- A way to generate the OFDM signal:
  - Modulate the sub-carriers, using for this purpose *K* modulators QAM (or PSK, QPSK, ...)



OFDM Implementation

- Implementation of OFDM - Immediate solution:
  - Modulator for each OFDM sub-carrier



$$x_K(t) = I_K(t).\cos(2\pi f_p t) - Q_K(t).\sin(2\pi f_p t),\ k = 0, 1, ..., K - 1$$

OFDM Implementation

- Implementation of OFDM - Immediate solution:
  - In practice K (number of sub-carriers) can reach thousands
  - K emitters with the above structure + K corresponding receivers: implementation not feasible
  - It is necessary to find an alternative feasible method

- Implementation of OFDM - Practical solution with DSP
  - Using IDFT (Inverse Discrete Fourier Transform) on the transmitter and DFT on the receiver

DFT Review

- The DFT is itself a <u>sequence</u> rather than a function of a continuous variable, and it corresponds to samples, equally spaced in frequency, of the <u>DTFT</u> of the signal
- In addition to its theoretical importance as a Fourier representation of sequences, the DFT plays a central role in the implementation of a variety of digital signal-processing algorithms
- This is because efficient algorithms exist for the computation of the DFT

DFT Review

- The Fourier series representation of the periodic sequence corresponds to the DFT of the finite-length sequence
- The approach is to define the Fourier series representation for periodic sequences and to study the properties of such representations
- Then, we repeat essentially the same derivations, assuming that the sequence to be represented is a finite-length sequence

DFT Review

- The Fourier series representation of a continuous-time periodic signal generally requires infinitely many harmonically related complex exponentials,
  - whereas the Fourier series for any discrete-time signal with period N requires only N harmonically related complex exponentials
- When the Fourier series is used to represent finite-length sequences, it is called the Discrete Fourier Transform or DFT

DFT Review

- In developing, discussing, and applying the DFT, it is always important to remember that the representation through samples of the Fourier Transform is in effect a representation of the finite-duration sequence by a periodic sequence, one period of which is the finite duration sequence that we wish to represent

DFT Review

- DFT important relations:

$$x[n] \overset{\mathcal{DFT}}{\longleftrightarrow} X[k]$$

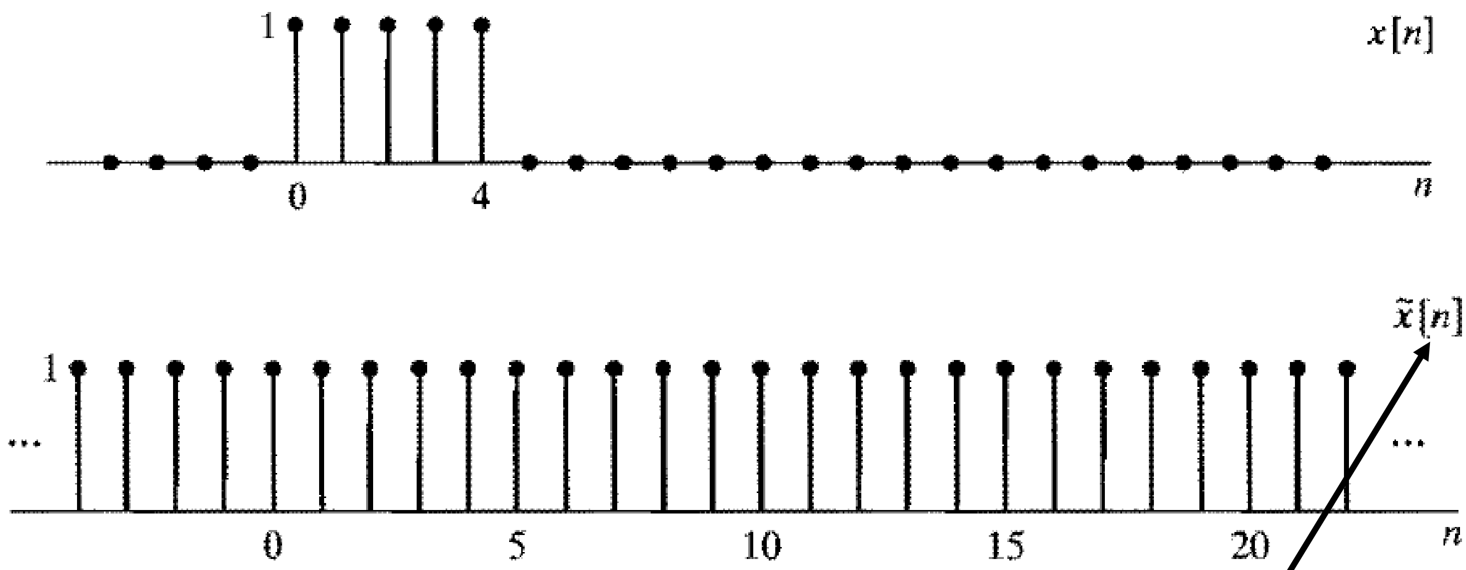
$$\text{Analysis equation:} \quad X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \qquad 0 \le k \le N-1,$$

$$\text{Synthesis equation:} \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \qquad 0 \le n \le N-1.$$

$$W_N = e^{-j(2\pi/N)}$$

DFT Review

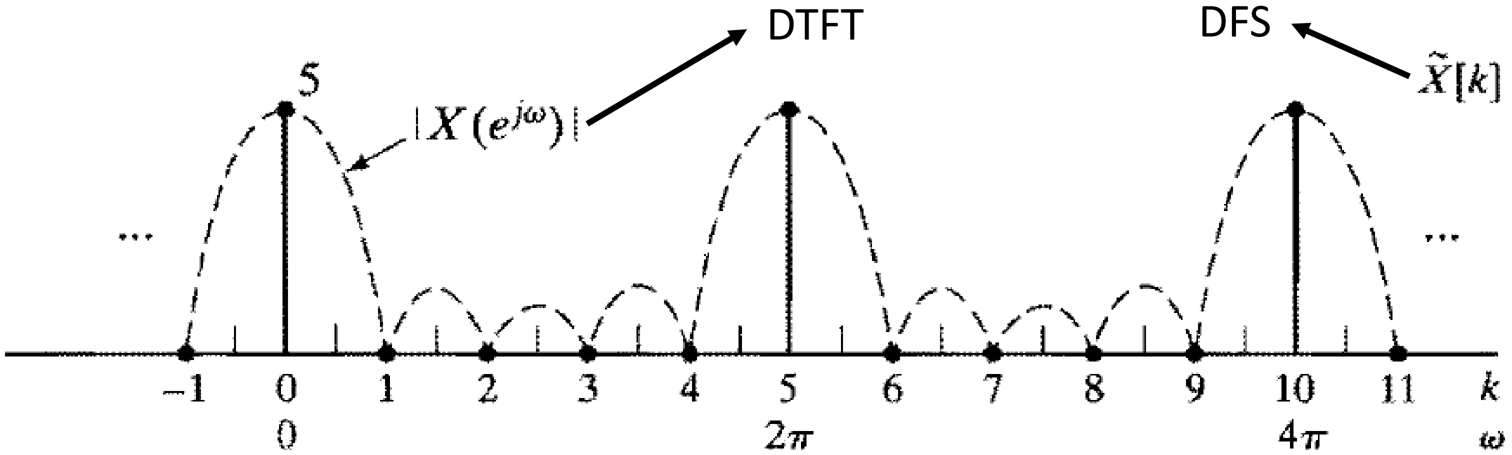
- The DFT of a Rectangular Pulse



Periodic sequence $\tilde{x}[n]$ formed from $x[n]$ with period $N = 5$

DFT Review

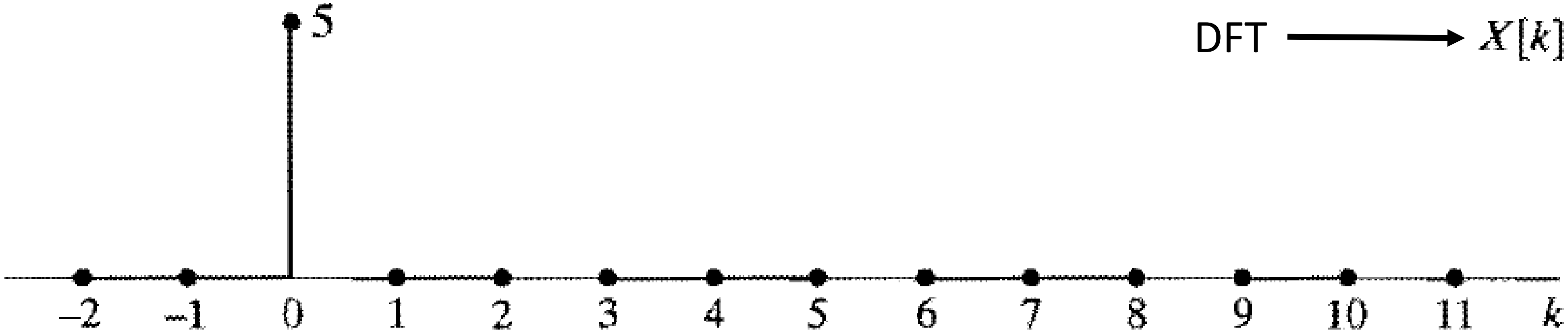
- The DFT of a Rectangular Pulse



$$\tilde{X}[k] = \sum_{n=0}^{4} e^{-j(2\pi k/5)n} = \frac{1 - e^{-j2\pi k}}{1 - e^{-j(2\pi k/5)}} = \begin{cases} 5, & k = 0, \pm 5, \pm 10, \ldots, \\ 0, & \text{otherwise}; \end{cases}$$

DFT Review

- The DFT of a Rectangular Pulse



OFDM Implementation

- Implementation of OFDM - Practical solution with DSP
  - The following mathematical description effectively corresponds to how the signal is implemented in practical systems by digital signal processing (DSP)
  - Each f_k sub-carrier is modulated in PSK, QPSK, QAM
  - It is given by:

$$x_K(t) = I_K(t).\cos(2\pi f_p t) - Q_K(t).\sin(2\pi f_p t),\ k = 0, 1, ..., K - 1$$

OFDM Implementation

- Implementation of OFDM - Practical solution with DSP
- Defining now:

$$X_K(t) = I_K(t) + j.Q_K(t)$$

$$x_K(t) = Re\,[X_K(t).e^{j2\pi f_k t}\,]$$

- The OFDM signal is the sum of the *K* sub-carriers:

$$x_p(t) = \sum_{k=0}^{K-1} x_k(t) = Re\left[\sum_{k=0}^{K-1} X_k(t) e^{j2\pi f_k t}\right]$$

OFDM Implementation

- Implementation of OFDM - Practical solution with DSP
- Defining *w*(*t*) as:

$$w(t) = \sum_{k=0}^{K-1} X_k(t) e^{j2\pi f_k t}$$

- To obtain OFDM signal:

$$x_p(t) = Re[w(t)]$$

OFDM Implementation

- Implementation of OFDM - Practical solution with DSP
- Using IDFT (Inverse Discrete Fourier Transform)
  - For this purpose, $w(t)$ samples at nT/K moments are considered:

$$w(t) = \sum_{k=0}^{K-1} X_k(t) e^{j2\pi f_k t}$$

$$w\left(n\frac{T}{K}\right) = \sum_{k=0}^{K-1} X_k\left(n\frac{T}{K}\right) e^{j2\pi f_k n\frac{T}{K}}$$

$n = 0, 1, \ldots, K - 1$

T - duration of the symbol on each subcarrier

K - number of subcarriers

OFDM Implementation

- Implementation of OFDM - Practical solution with DSP
- Considering: $f_0 = 0$, then $f_k = k.\Delta f = k/T$

$$w(n) = \sum_{k=0}^{K-1} X_k(n) e^{j2\pi \frac{k}{T} n \frac{T}{K}} = \sum_{k=0}^{K-1} X_k(n) e^{j2\pi k \frac{n}{K}}, \; n = 0, 1, \; \ldots \; , K-1$$

- Except for a multiplying constant (1/*N*), the above formula is the equation of an *N*-point Inverse Discrete Fourier transform (IDFT)

OFDM Implementation

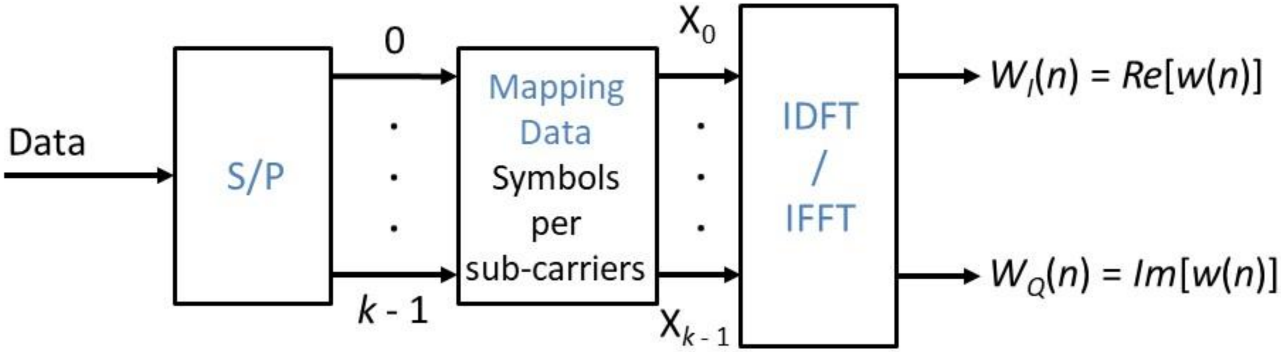
- Implementation of OFDM - Practical solution with DSP

$$w(n) = \sum_{k=0}^{K-1} X_k(n) e^{j2\pi \frac{k}{T} n \frac{T}{K}} = \sum_{k=0}^{K-1} X_k(n) e^{j2\pi k \frac{n}{K}}, \; n = 0, 1, \ldots, K-1$$

- **Conclusion:**
  - $w(n)$ = IDFT [$X_K(n)$] -> Practical Method to implement an OFDM T_x

OFDM Implementation

- The following figure shows the front-end of an OFDM transmitter:

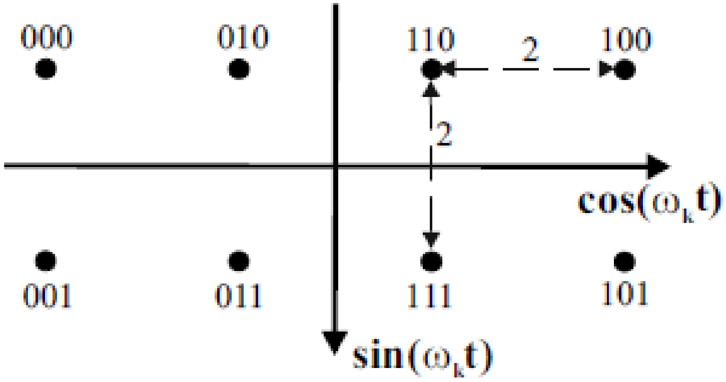


  – Let's suppose that 4 subcarriers (K = 4) are used, each modulated according to the following expression:

$$x_K(t) = I_K(t).\cos(2\pi f_k t) - Q_K(t).\sin(2\pi f_k t),\ k = 0, 1, ..., K - 1$$

OFDM Implementation

- The mapping of the input bits in each subcarrier is indicated in the figure constellation:

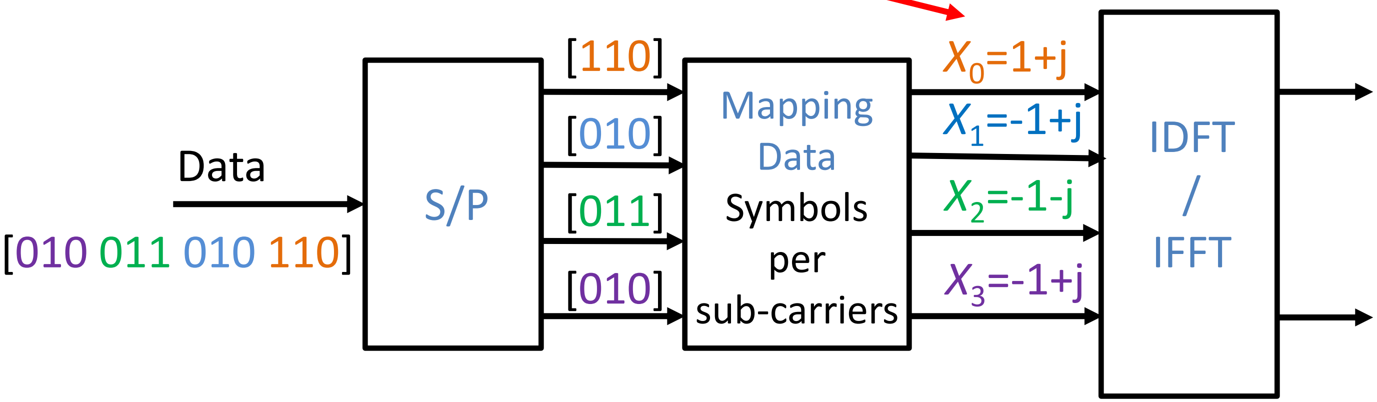


  - The input data is the following binary sequence [010 011 010 110]
  - For the purposes, consider that the first bit to reach the transmitter is the rightmost bit in the sequence and that the 3 rightmost bits are sent to the upper branch of the series-parallel converter

OFDM Implementation

- Also consider that the bits indicated in the constellation obey the same order (the rightmost bit is the first to arrive)
- **1.** Determine the values of all X_k:

$$X_K(t) = I_K(t) + j.Q_K(t)$$



OFDM Implementation

- Also consider that the bits indicated in the constellation obey the same order (the rightmost bit is the first to arrive)
- **2.** Determine (without using programs that implement the IFFT) the first two sample values for each of the outputs on the front of a transmitter:

$$w(n) = \frac{1}{K}\sum_{k=0}^{K-1} X_k e^{j2\pi k\frac{n}{K}}, \quad n = 0, 1, \ldots, K-1$$

- Then:

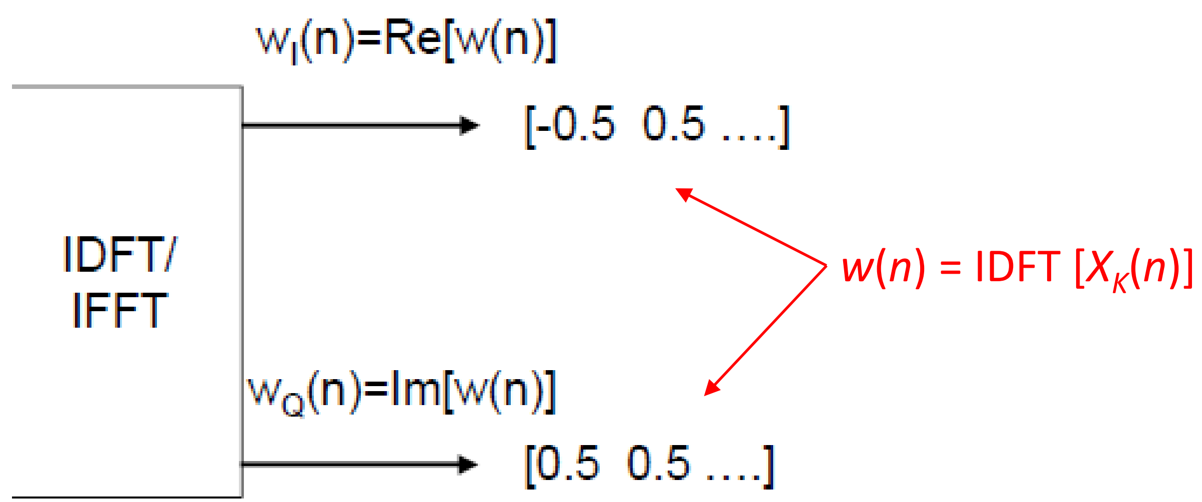
$$w(0) = \frac{1}{K}\sum_{k=0}^{3} X_k = \frac{(-2+2j)}{4} = -0.5+0.5j$$

OFDM Implementation

- And:

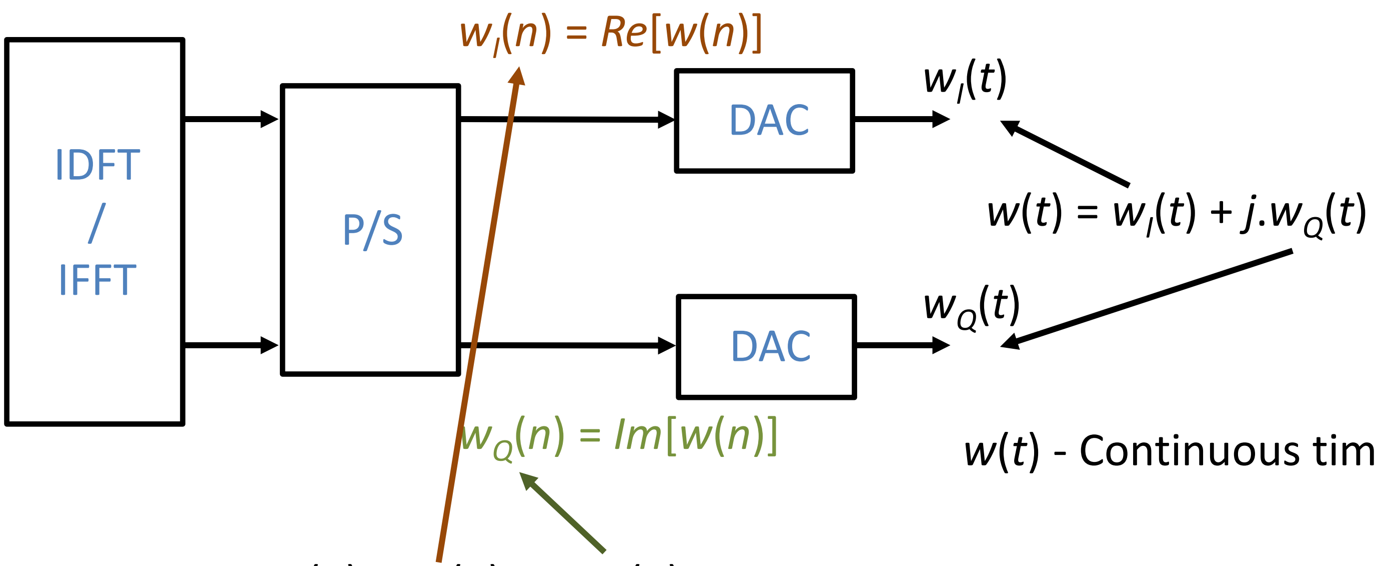
$$w(1)=\frac{1}{4}\sum_{k=0}^{3}X_k e^{j2\pi k\frac{1}{K}}=\frac{1}{4}\left(X_0+X_1e^{j2\pi\frac{1}{4}}+X_2e^{j2\pi\frac{2}{4}}+X_3e^{j2\pi\frac{3}{4}}\right)=\frac{1}{4}\left(X_0+X_1e^{j\frac{\pi}{2}}+X_2e^{j\pi}+X_3e^{j3\frac{\pi}{2}}\right)$$

$$=\frac{1}{4}\left((1+j)+(-1+j)*j-(-1-j)-(-1+j)*j\right)=\frac{1}{4}(2+2j)=0.5+0.5j$$



OFDM Implementation

- Continuous time *w*(*t*) signal generation using DACs



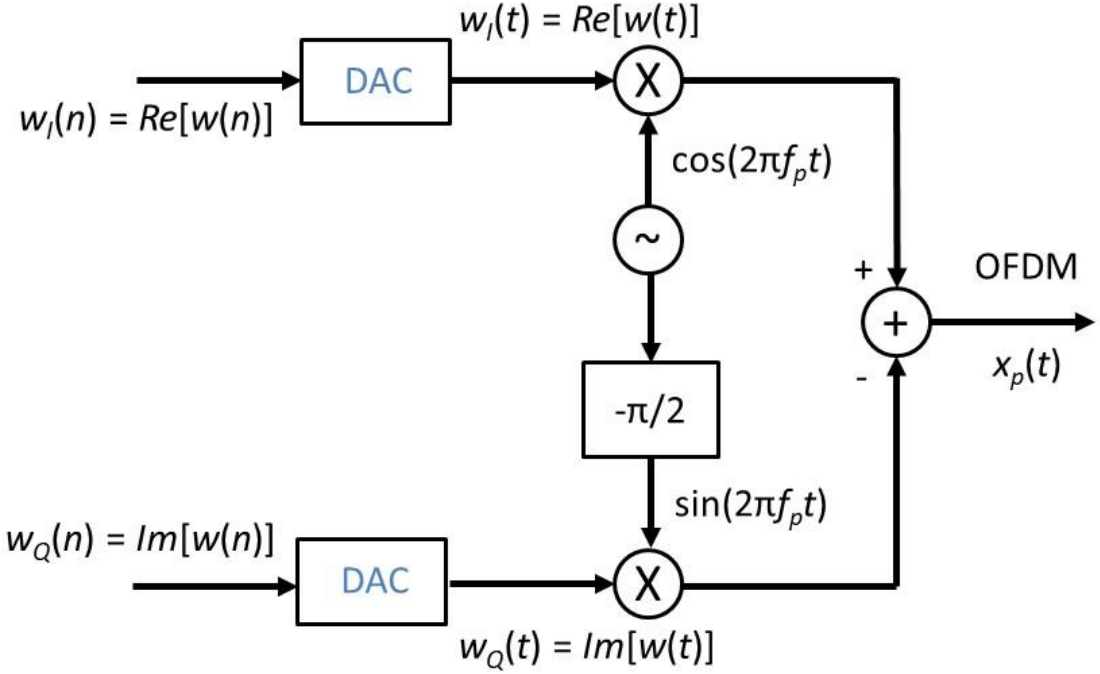
w(*t*) - Continuous time version of *w*(*n*)

Samples of *w*(*t*): $w(n) = w_I(n) + j.w_Q(n)$

w(*n*) - Samples of the composite signal (sum of all modulated sub-carriers)

OFDM Implementation

- UP conversion to the carrier frequency



$$x_p(t) = w_I(t)\cos\left(2\pi f_p t\right) - w_Q(t)\,sen\left(2\pi f_p t\right)$$

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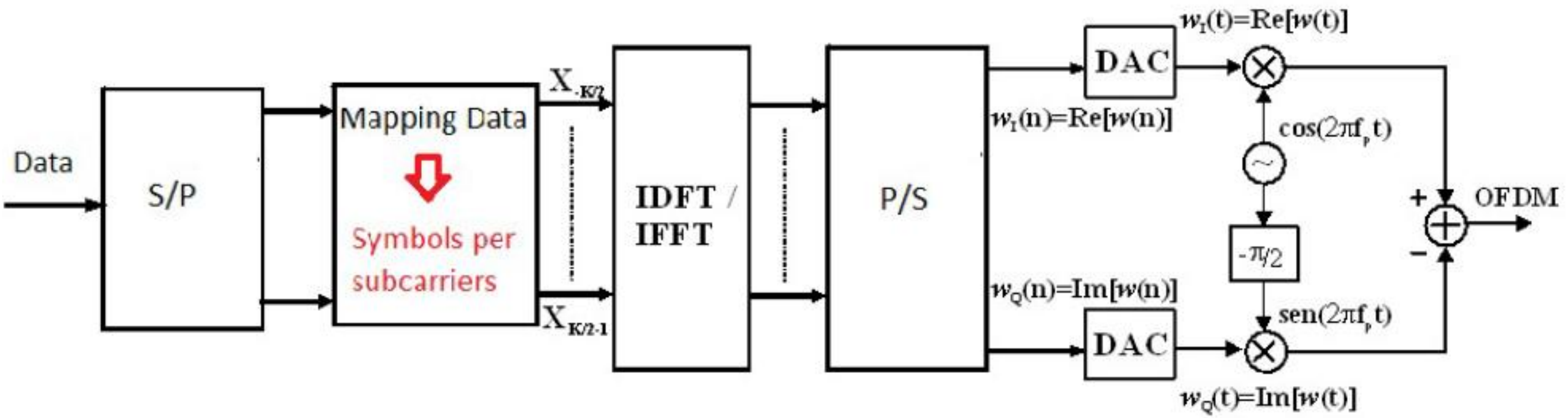
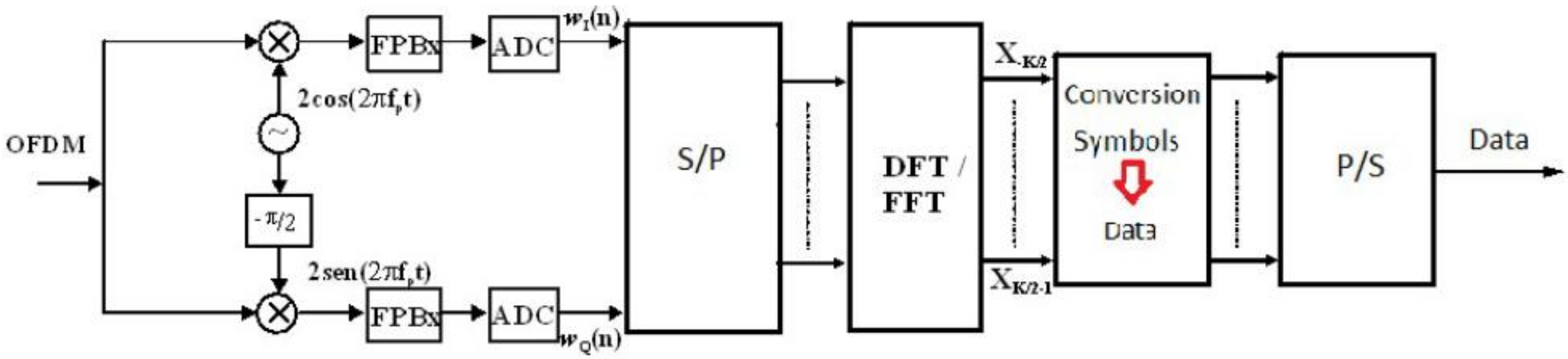
OFDM Implementation

- OFDM Transmitter



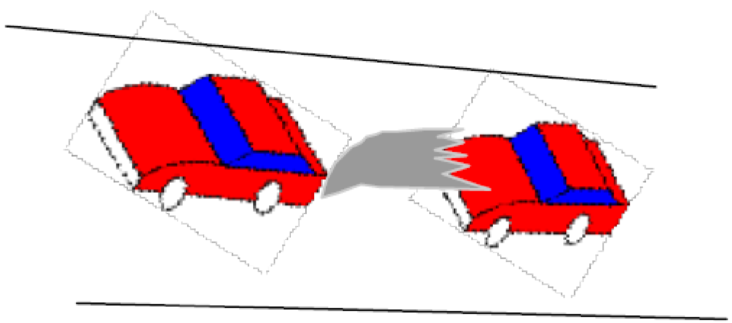
OFDM Implementation

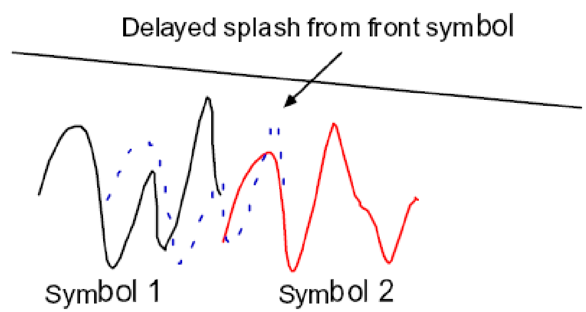
- OFDM Receiver



OFDM with guard interval (cyclic extension)

- Delay spread - use of cyclic prefix to mitigate it
- You are driving in rain, and the car in front splashes a bunch of water on you. What do you do?
  - You move further back, you put a little distance between you and the front car, far enough so that the splash won't reach you
  - If we equate the reach of splash to delay spread of a splashed signal, then we have a better picture of the phenomena and how to avoid it



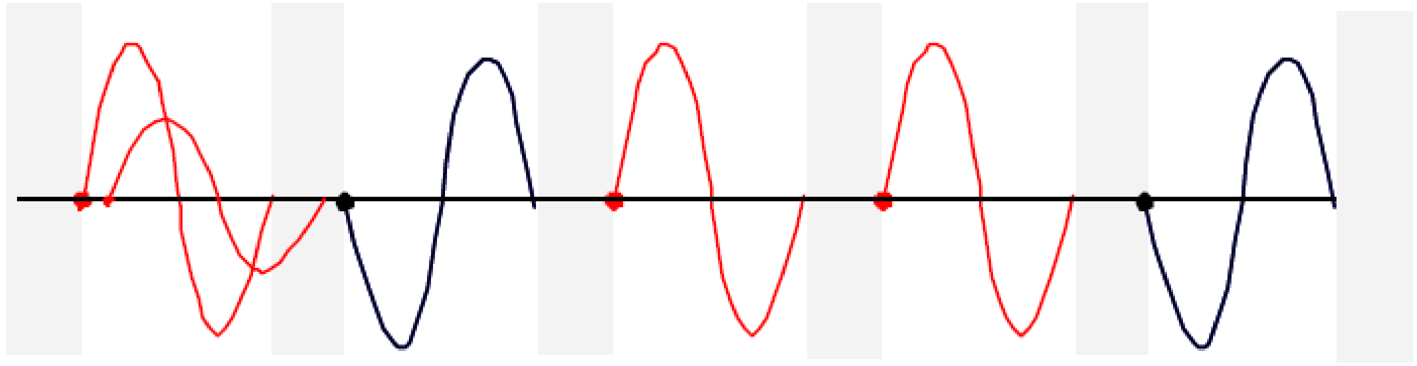


OFDM with guard interval (cyclic extension)

- Delay spread - use of cyclic prefix to mitigate it
- Delay spread is like the undesired splash you might get from the car ahead of you
  - In fading, the front symbol similarly throws a splash backwards which we wish to avoid.
- Increase distance from car in front to avoid splash
  - The reach of splash is the same as the delay spread of a signal
  - In composite, these splashes become noise and affect the beginning of the next symbol causing Inter-symbol interference

OFDM with guard interval (cyclic extension)

- Delay spread - use of cyclic prefix to mitigate it
- To mitigate this noise at the front of the symbol, we will move our symbol further away from the region of delay spread
- A little bit of blank space has been added between symbols to avoid the delay spread

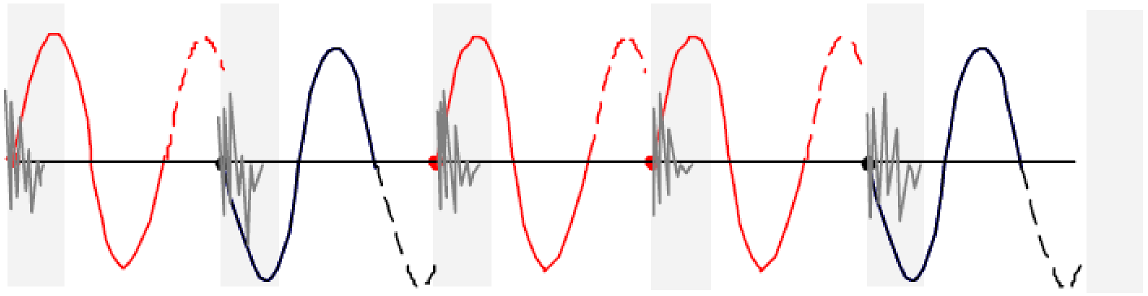


OFDM with guard interval (cyclic extension)

- Delay spread - use of cyclic prefix to mitigate it
- But we can not have blank spaces in signals
- This won't work for the hardware which likes to output signals continuously
- So, it is clear we need to have something there
- Why don't we just let the symbol run longer as a first choice?
- If we just extend the symbol, then the begin of the symbol which is important since it allows figuring out what the phase of this symbol is, is now corrupted by the "splash"

OFDM with guard interval (cyclic extension)

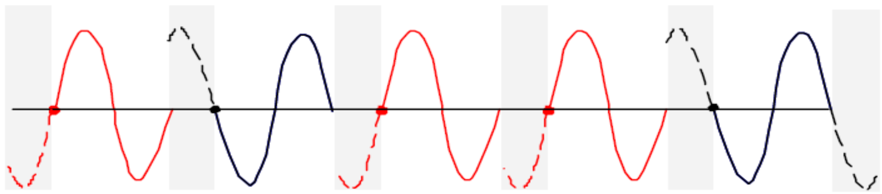
- Delay spread - use of cyclic prefix to mitigate it



- We extend the symbol into the empty space, so the actual symbol is more than one cycle
- Now the start of the symbol is still in the danger zone, and this start is the most important thing about our symbol since the slicer needs it in order to make a decision about the bit

OFDM with guard interval (cyclic extension)

- Delay spread - use of cyclic prefix to mitigate it
- We don't want the start of the symbol to fall in this region, so lets just slide the symbol backwards, so that the start of the original symbol lands at the outside of this zone. And then fill this area with something
- Slide the symbol to start at the edge of the delay spread time and then fill the guard space with a copy of what turns out to be tail end of the symbol



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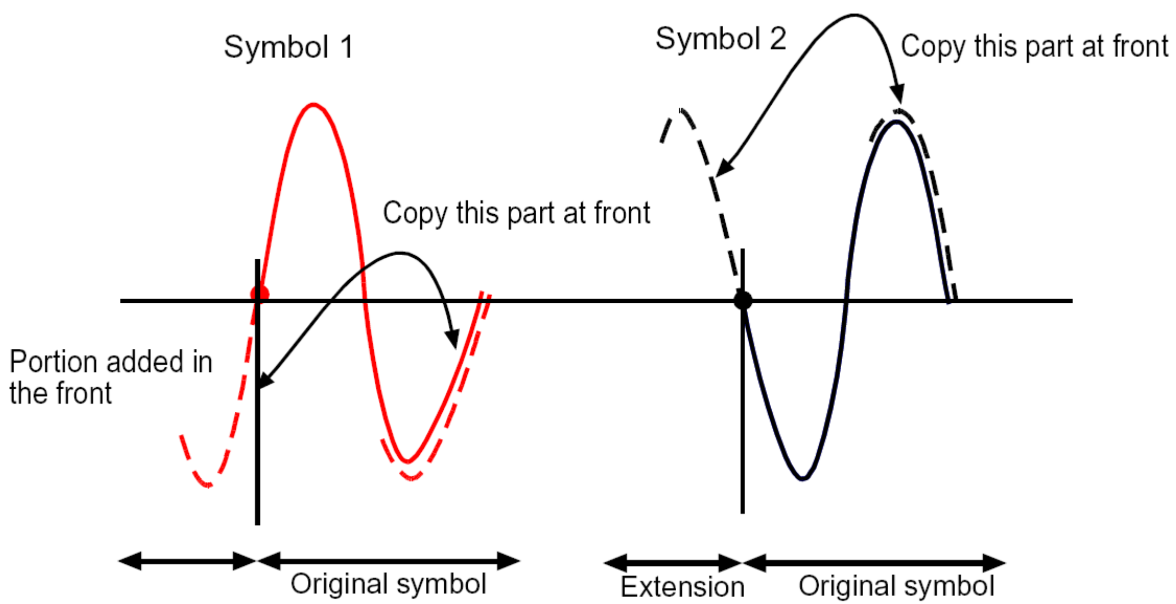
OFDM with guard interval (cyclic extension)

- Delay spread - use of cyclic prefix to mitigate it
- 1. We want the start of the symbol to be out of the delay spread zone, so it is not corrupted
- and
- 2. We start the signal at the new boundary such that the actual symbol edge falls outside this zone
- We will be extending the symbol so it is 1.25 times as long, to do this, copy the back of the symbol and glue it in the front
- In reality, the symbol source is continuous, so all we are doing is adjusting the starting phase and making the symbol period longer

OFDM with guard interval (cyclic extension)

- Delay spread - use of cyclic prefix to mitigate it
- All books talk about it as a copy of the tail end. And the reason is that in digital signal processing, we do it this way



OFDM with guard interval (cyclic extension)

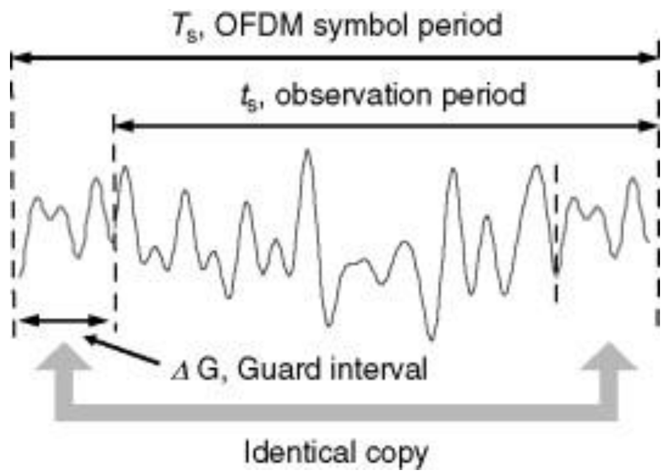
- Delay spread - use of cyclic prefix to mitigate it
- This procedure is called adding a cyclic prefix
- Since OFDM, has a lot of carriers, we would do this to each and every carrier
- But that’s only in theory
- In reality since the OFDM signal is a linear combination, we can add cyclic prefix just once to the composite OFDM signal
- The prefix is anywhere from 10% to 25% of the symbol time

OFDM with guard interval (cyclic extension)

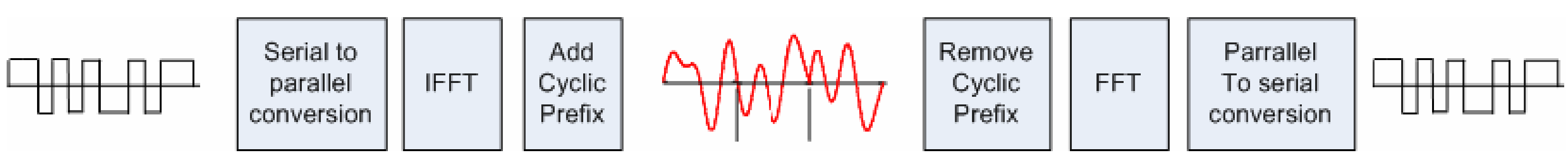
- Delay spread - use of cyclic prefix to mitigate it
- The whole process can be done only once to the OFDM signal, rather than doing it to each and every sub-carrier
- We add the prefix after doing the IFFT just once to the composite signal
- After the signal has arrived at the receiver, first remove this prefix, to get back the perfectly periodic signal so it can be FFT'd to get back the symbols on each carrier
- However, the addition of cyclic prefix which mitigates the effects of link fading and inter symbol interference, increases the bandwidth

OFDM with guard interval (cyclic extension)

- Delay spread - use of cyclic prefix to mitigate it



- Addition of Cyclic Prefix to the OFDM signal further improves its ability to deal with fading and interference



Advantages of OFDM

- Simple solution for the multipath problem
- The FFT allows efficient implementations
- High spectral efficiency
- Supports multiple access schemes
- Each carrier can be modulated using different schemes
- Each carrier can be adapted to different SNR conditions

Disadvantages of OFDM

- Large number of sub-carriers with possible variations in instantaneous transmitted power (QAM signal)
- Variations in the total instantaneous power of the OFDM signal even higher peak-to-average power ratio (PAPR)
- This leads to reduced efficiency in power amplifiers, a particularly important problem in mobile terminals

Applications

- Wireless LANs
  - IEEE 802.11
- DAB and DVB
- xDSL (multitone)
- 3GPP Long Term Evolution (LTE) (HSOPA)
- Wimax
- Underwater communications with ultrasounds

Reading Material

- Henrik Schulze and Christian Luders, Theory and Applications of OFDM and CDMA, Wideband Wireless Communications, John Wiley & Sons Inc., 2005 (ISBN-13 978-0-470-85069-5)
- Monteiro, P., University of Aveiro, Slides of Digital Communications Systems, Part II, Multi-Carrier Digital Transmission
- Simon Haykin, Communication Systems, McMaster Univ, 5th Edition, 2009 (ISBN 978-047-16979-0-9)

Reading Material